

PROGRAMMING FUNDAMENTALS

LABORATORY ASSIGNMENT — 02

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Section: D

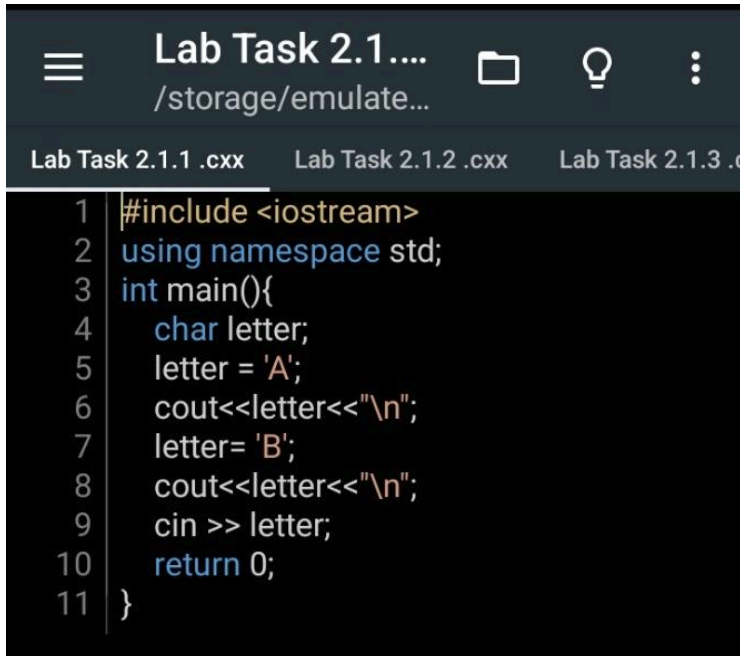
Department: BS Computer Science

Subject: Programming Fundamentals

Lab: PF Lab 2

TASK 1 — OBSERVATION OF PROGRAMS

Program 1



```
1 #include <iostream>
2 using namespace std;
3 int main(){
4     char letter;
5     letter = 'A';
6     cout<<letter<<"\n";
7     letter= 'B';
8     cout<<letter<<"\n";
9     cin >> letter;
10    return 0;
11 }
```

Output:



```
← TAB _ ⋮
A
B
█
```

Observation:

The variable letter is declared as a char. First, the character 'A' is assigned to it and displayed. Then, 'B' is assigned and displayed.

TASK-1 (Observation)

Program-1:

```
#include <iostream>
using namespace std;
int main(){
    char letter;
    letter = 'A';
    cout << letter << "\n";
    letter = 'B';
    cout << letter << "\n";
    cin >> letter;
    return 0;
}
```

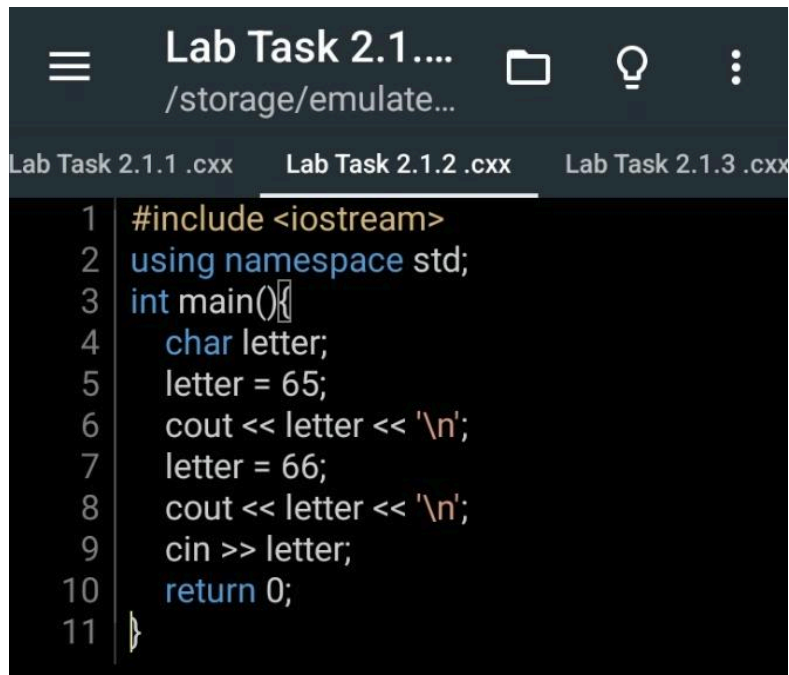
Output

A
B

Observation:

The variable letter is declared as a char. First, the character 'A' is assigned to it and displayed. Then, 'B' is assigned and displayed.

Program 2



```
1 #include <iostream>
2 using namespace std;
3 int main(){
4     char letter;
5     letter = 65;
6     cout << letter << '\n';
7     letter = 66;
8     cout << letter << '\n';
9     cin >> letter;
10    return 0;
11 }
```

Output:



```
A
B
```

Observation:

The variable letter is declared as a char. The values 65 and 66 are assigned to it. According to ASCII, 65 represents A and 66 represents B.

Program- 2:

```
#include <iostream>
using namespace std;
int main(){
    char letter;
    letter = 65;
    cout << letter << "\n";
    letter = 66;
    cout << letter << "\n";
    cin >> letter;
    return 0;
}
```

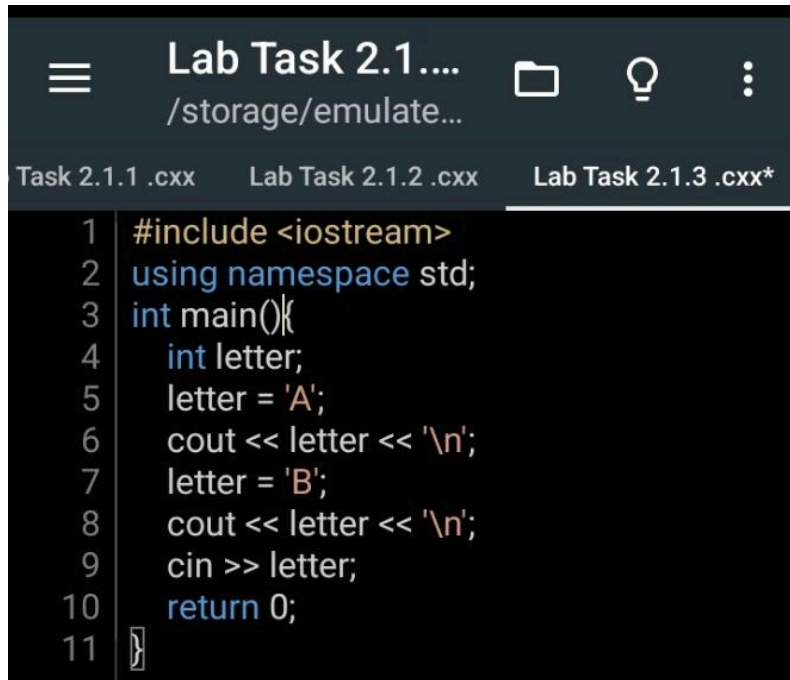
Output

A
B

Observation:

The variable letter is declared as a char. The values 65 and 66 are assigned to it. According to ASCII, 65 represents A and 66 represents B.

Program 3



```
1 #include <iostream>
2 using namespace std;
3 int main(){
4     int letter;
5     letter = 'A';
6     cout << letter << '\n';
7     letter = 'B';
8     cout << letter << '\n';
9     cin >> letter;
10    return 0;
11 }
```

Output:



```
65
66
```

Observation:

The variable letter is declared as an int. The characters 'A' and 'B' are assigned to it. Their ASCII values are stored as integers.

Conclusion

ASCII assigns numerical codes to characters. For example, A has ASCII code 65 and B has ASCII code 66.

Program- 3 :

```
#include <iostream>
using namespace std;
int main(){
    int letter;
    letter = 'A';
    cout << letter << "\n";
    letter = 'B';
    cout << letter << "\n";
    cin >> letter;
    return 0;
}
```

Output

65

66

Observation: The variable letter is declared as an int. The characters 'A' and 'B' are assigned to it. Their ASCII values are stored as integers.

Conclusion: ASCII assigns numerical codes to characters. For Example, A has ASCII code 65 and B has ASCII code 66.

Task 2 — Arithmetic Operations

Take two integers as input from the user and perform addition, subtraction, multiplication, and division. Then display the results.

Program:

```
≡ Lab Task 2.2... /storage/emulate...  
1 #include <iostream>  
2 using namespace std;  
3 int main(){  
4 int a,b;  
5 cout<<"Enter first number";  
6 cin>>a;  
7 cout<<"Enter second number";  
8 cin>>b;  
9 cout<<"Addition="<<a+b<<endl;  
10 cout<<"Subtraction="<<a-b<<endl;  
11 cout<<"Multiplication="<<a*b<<endl;  
12 cout<<"Division="<<a/b|  
13 return 0;  
14 }
```

Output:

```
← TAB _ ⋮  
Enter first number10  
Enter second number5  
Addition=15  
Subtraction=5  
Multiplication=50  
Division=2  
[Program finished]
```

Task-2 (Arithmetic Operations)

Take two integers as input from the user and perform addition, subtraction, multiplication and division. Then display the results.

Program:

```
#include <iostream>
using namespace std;
int main(){
    int a, b;
    cout << "Enter first number: ";
    cin >> a;
    cout << "Enter second number: ";
    cin >> b;
    cout << "Addition=" << a+b << endl;
    cout << "Subtraction=" << a-b << endl;
    cout << "Multiplication=" << a*b << endl;
    cout << "Division=" << a/b;
    return 0;
}
```

Output

```
Enter first number: 10
Enter second number: 5
Addition = 15
Subtraction = 5
Multiplication = 50
Division = 2
```

Task 3 — Output Prediction

Write the output of the given C++ program:

```
1 #include "stdafx.h"
2 #include <iostream>
3 using namespace std;
4 int main(){
5     int val1;
6     int val2;
7     int val3;
8     cout<<"Please enter your 3 numbers:";
9     cin>>val1>>val2>>val3;
10    cout<<val1<<"\n"<<val2<<"\n"<<val3<<"\n";
11    cout<<val3<<"\n"<<val2<<"\n"<<val1<<"\n";
12    system("PAUSE");
13    return 0;
14 }
```

Output:

```
← TAB _ ⋮
Please enter your 3 numbers:100
200
300
100
200
300
300
200
100
sh: PAUSE: not found

[Program finished]
```

(Note: "system("PAUSE")" is not supported on Android/Cxxdroid.)

Task - 3 (Output)

Program:

```
#include <iostream>
using namespace std;
int main(){
    int val1;
    int val2;
    int val3;
    cout << "Please enter 3 numbers:";
    cin >> val1 >> val2 >> val3;
    cout << val1 << "\n" << val2 << "\n" << val3
    << "\n";
    cout << val3 << "\n" << val2 << "\n" << val1
    << "\n";
    system("PAUSE");
    return 0;
}
```

Output

Please enter 3 numbers: 100

200

300

100

200

300

300

200

100

sh: PAUSE: not found

NOTE

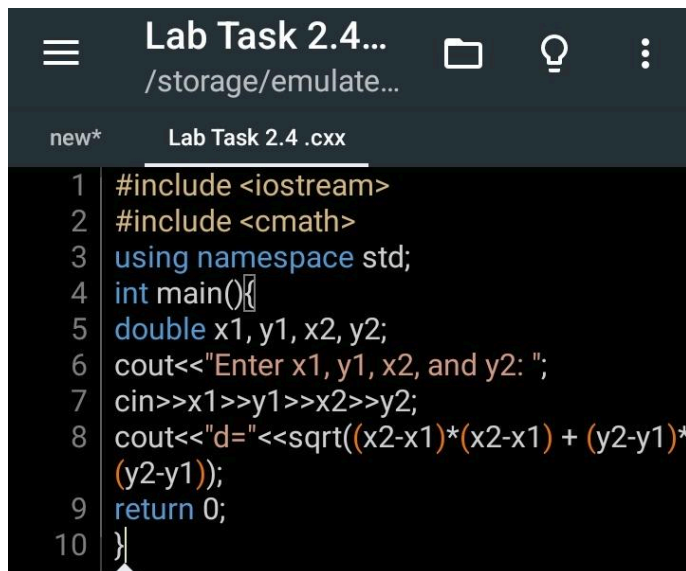
"system("PAUSE")"
is not supported
on Adroid/Cxx-
-droid.

Task 4 — Distance Between Two Points

Write a C++ program to calculate the distance between two points. The points are "(x1, y1)" and "(x2, y2)", where "x1", "y1", "x2", and "y2" are double values.

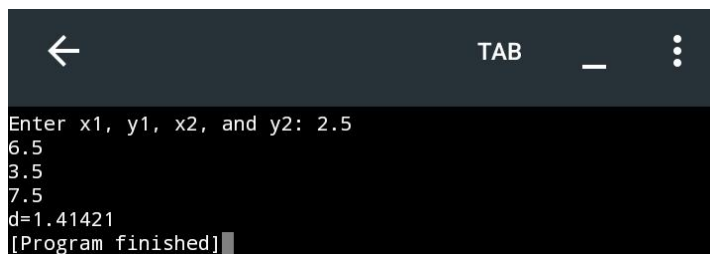
$$d = \sqrt{(x2 - x1)^2 + (y2 - y1)^2}$$

Program:



```
Lab Task 2.4...  
/storage/emulate...  
new* Lab Task 2.4 .cxx  
1 #include <iostream>  
2 #include <cmath>  
3 using namespace std;  
4 int main(){  
5     double x1, y1, x2, y2;  
6     cout<<"Enter x1, y1, x2, and y2: ";  
7     cin>>x1>>y1>>x2>>y2;  
8     cout<<"d="<<sqrt((x2-x1)*(x2-x1) + (y2-y1)*  
9         (y2-y1));  
10    return 0;  
}
```

Output:



```
← TAB _ ⋮  
Enter x1, y1, x2, and y2: 2.5  
6.5  
3.5  
7.5  
d=1.41421  
[Program finished]
```

Input:

x1=2.5, x2=3.5, y1=6.5, y2=7.5

Output:

d=1.41421

TASK - 4 (Calculate Distance)

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Program:

```
#include <iostream>
using namespace std;
int main(){
    double x1, y1, x2, y2;
    cout << "Enter x1, y1, x2 and y2: ";
    cin >> x1 >> y1 >> x2 >> y2;
    cout << "d=" << sqrt((x2-x1)*(x2-x1) + (y2-y1)
        *(y2-y1));
    return 0;
}
```

Output

```
Enter x1, y1, x2 and y2: 2.5
6.5
3.5
7.5
d= 1.41421
```

Input:

$x_1 = 2.5$, $y_1 = 6.5$

$x_2 = 3.5$, $y_2 = 7.5$

Output:

$d = 1.41421$

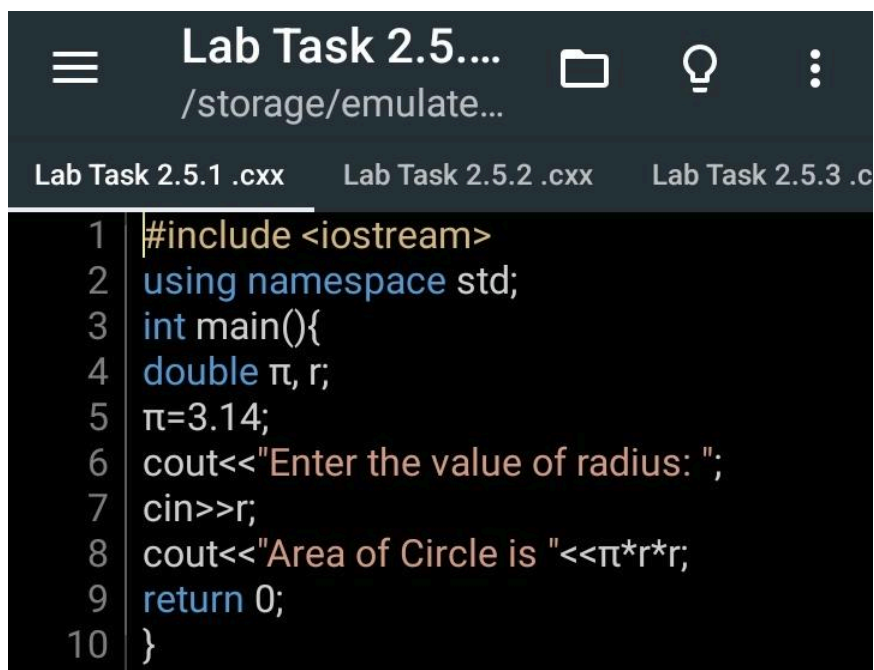
Task 5 — Find The Areas

Question:

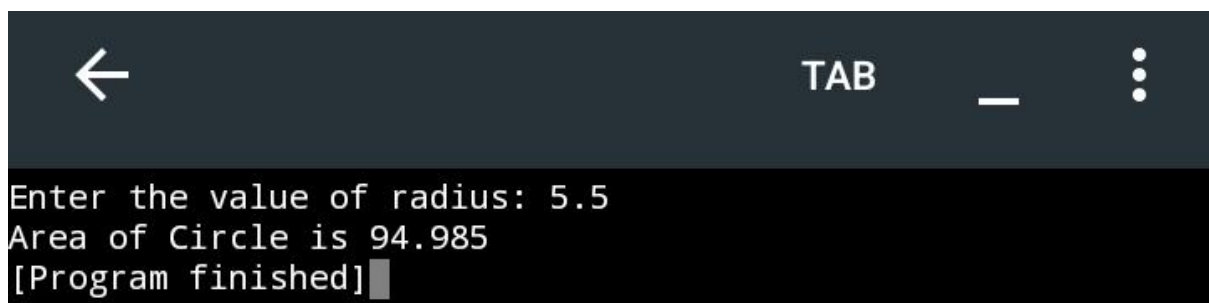
Write separate C++ programs to calculate:

- a) Area of a Circle
- b) Area of a Rectangle
- c) Area of a Triangle

a) Area of a Circle



```
Lab Task 2.5.1 .cxx  Lab Task 2.5.2 .cxx  Lab Task 2.5.3 .c
1  #include <iostream>
2  using namespace std;
3  int main(){
4  double π, r;
5  π=3.14;
6  cout<<"Enter the value of radius: ";
7  cin>>r;
8  cout<<"Area of Circle is "<<π*r*r;
9  return 0;
10 }
```



```
←  TAB  _  ⋮
Enter the value of radius: 5.5
Area of Circle is 94.985
[Program finished]
```

Task - 5 (Find Areas)

(a) Area of Circle:

Write C++ program to calculate the area of a circle.

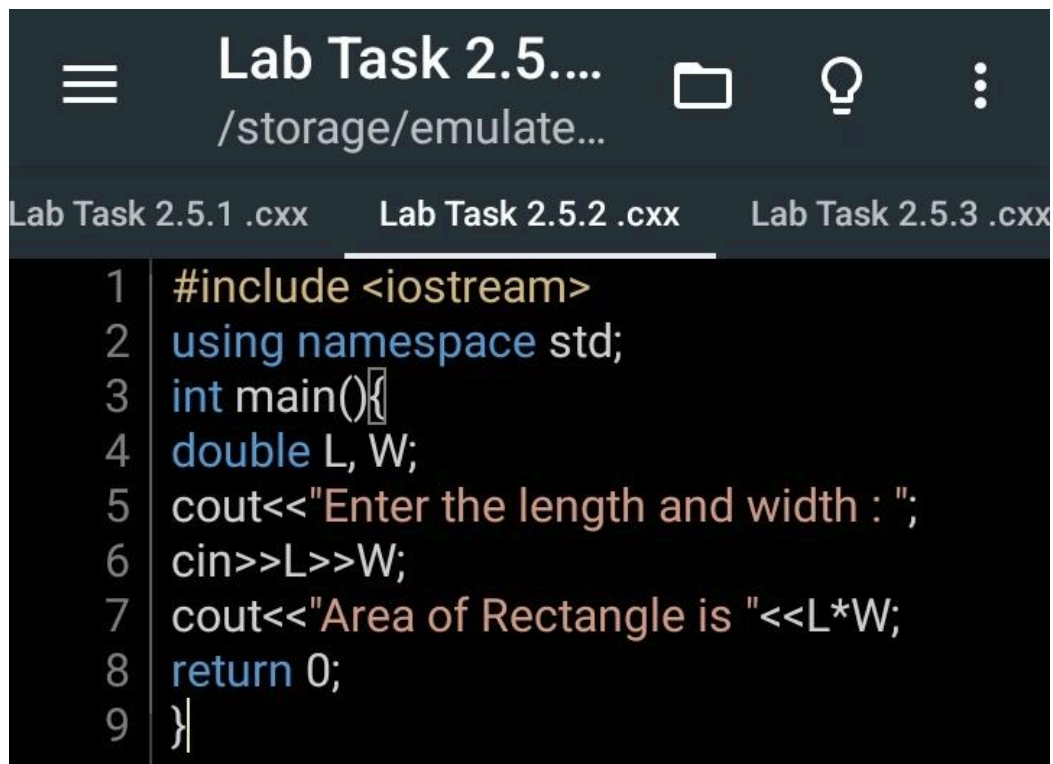
Program:

```
#include <iostream>
using namespace std;
int main() {
    double  $\pi$ , r;
     $\pi$  = 3.14;
    cout << "Enter value of radius: ";
    cin >> r;
    cout << "Area of circle is: " <<  $\pi$ *r*r;
    return 0;
}
```

Output

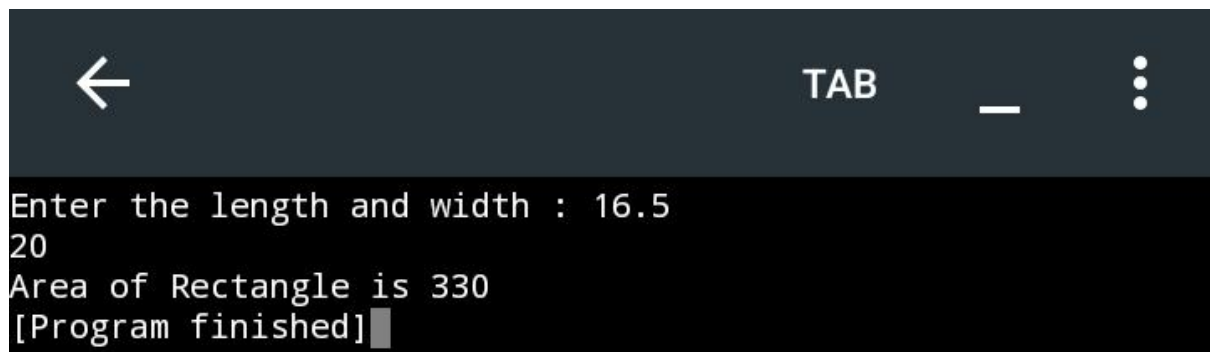
```
Enter value of radius: 5.5
Area of circle is: 94.985
```

b) Area of a Rectangle



The screenshot shows a code editor with a dark theme. The title bar at the top reads "Lab Task 2.5...." and the address bar shows "/storage/emulate...". There are three tabs open: "Lab Task 2.5.1 .cxx", "Lab Task 2.5.2 .cxx" (which is the active tab), and "Lab Task 2.5.3 .cxx". The code in the active tab is as follows:

```
1 #include <iostream>
2 using namespace std;
3 int main(){
4     double L, W;
5     cout<<"Enter the length and width : ";
6     cin>>L>>W;
7     cout<<"Area of Rectangle is "<<L*W;
8     return 0;
9 }
```



The screenshot shows a terminal window with a dark theme. The title bar at the top has a back arrow, the text "TAB", and a close button. The terminal output is as follows:

```
Enter the length and width : 16.5
20
Area of Rectangle is 330
[Program finished]
```

(b) Area of Rectangle:

Write a C++ program that calculates the Area of Rectangle.

Program:

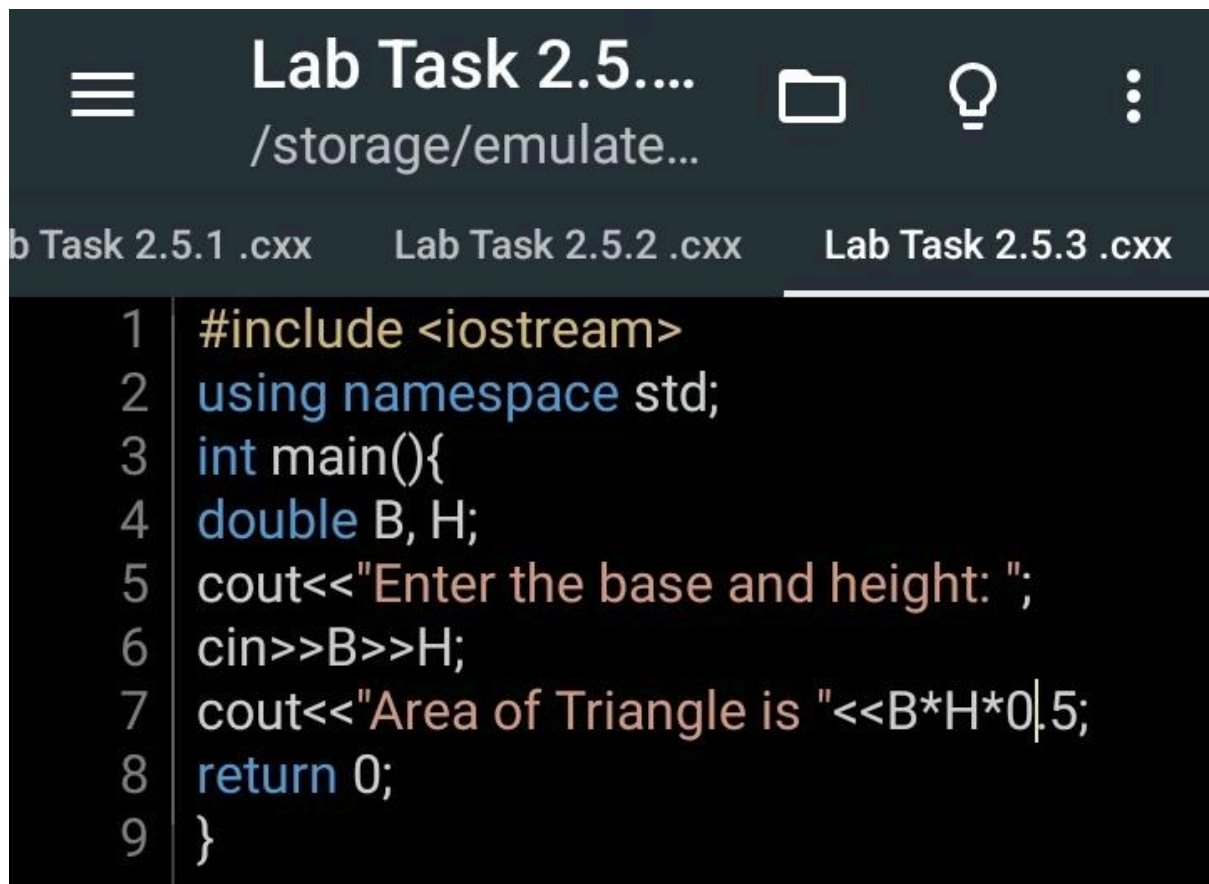
```
#include <iostream>
using namespace std;
int main(){
    double L, W;
    cout << "Enter length and width: ";
    cin >> L >> W;
    cout << "Area of Rectangle is " <<
    L * W;
    return 0;
}
```

Output

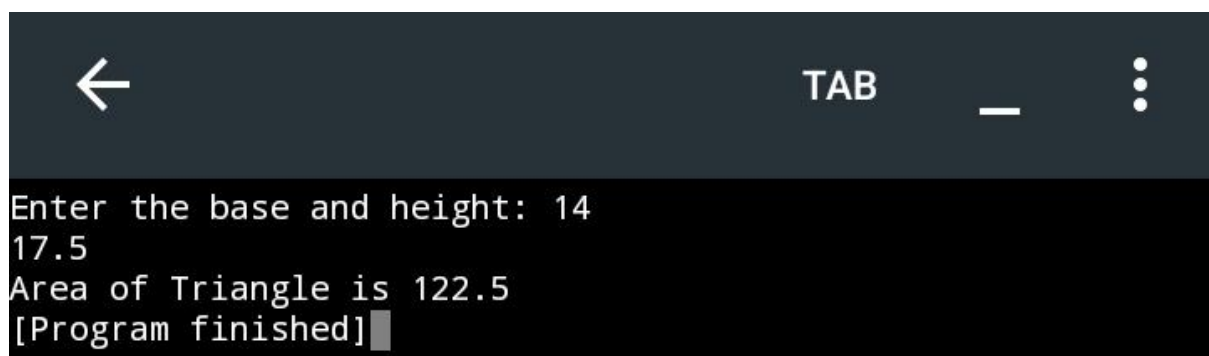
```
Enter length and width: 16.5
20
Area of Rectangle is 330
```

Program finished

c) Area of a Triangle



```
1 #include <iostream>
2 using namespace std;
3 int main(){
4     double B, H;
5     cout<<"Enter the base and height: ";
6     cin>>B>>H;
7     cout<<"Area of Triangle is "<<B*H*0.5;
8     return 0;
9 }
```



```
← TAB _ ⋮
Enter the base and height: 14
17.5
Area of Triangle is 122.5
[Program finished]
```

(C) Area of Triangle:

Write a program that calculates the Area of Triangle.

Program:

```
#include <iostream>
using namespace std;
int main(){
    double B, H;
    cout << "Enter Base and Height: ";
    cin >> B >> H;
    cout << "Area of Triangle is " <<
    B * H * 0.5;
    return 0;
}
```

Output

```
Enter Base and Height: 14
17.5
Area of Triangle is 122.5
```

Task 6 — Customer Purchase

A customer purchases five items. Store the prices of the five items in five variables. Display:

- Price of each item
- Subtotal
- Sales tax
- Total

Sales tax = 6%

Program

```
Lab Task 2.6...  
/storage/emulate...  
1 #include <iostream>  
2 using namespace std;  
3 int main()  
4 {  
5     double item1, item2, item3, item4, item5,  
6       Subtotal, Tax, SalesTax;  
7     item1= 12.95;  
8     item2= 24.95;  
9     item3=6.95;  
10    item4=14.95;  
11    item5=3.95;  
12    Tax=0.06;  
13    Subtotal= item1+item2+item3+item4+  
14    item5;  
15    SalesTax=Subtotal*Tax;  
16    cout<<"Price of item 1 = $"<<item1  
17    <<"\nPrice of item 2 = $"<<item2  
18    <<"\nPrice of item 3 = $"<<item3  
19    <<"\nPrice of item 4 = $"<<item4  
20    <<"\nPrice of item 5 = $"<<item5  
21    <<"\nThe Subtotal of the sale is "  
22    <<Subtotal  
23    <<"\nThe SalesTax is "<<SalesTax  
24    <<"\nTotal= "<<Subtotal+SalesTax;  
25    return 0;  
26 }
```

Output:

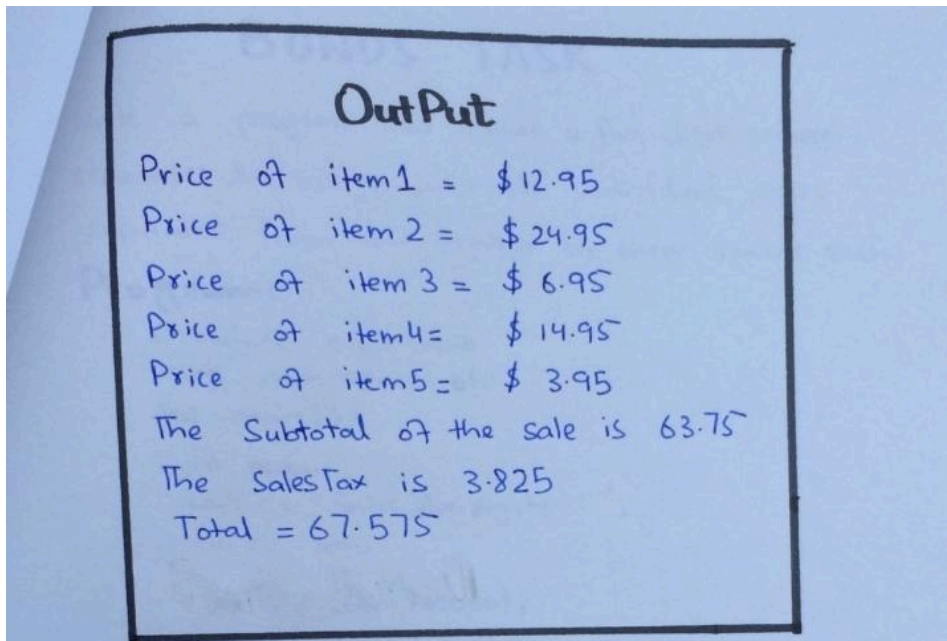
```
← TAB _ ⋮  
Price of item 1 = $12.95  
Price of item 2 = $24.95  
Price of item 3 = $6.95  
Price of item 4 = $14.95  
Price of item 5 = $3.95  
The Subtotal of the sale is 63.75  
The SalesTax is 3.825  
Total= 67.575  
[Program finished]
```

TASK - 6 (Customer Purchase)

A customer purchases five items. Store the prices in five variables. Display Price of each, Subtotal, Sales Tax and Total. Salestax = 6%

Program:

```
#include <iostream>
using namespace std;
int main() {
    double item1, item2, item3, item4, item5,
    Subtotal, Tax, SalesTax;
    item1 = 12.95;
    item2 = 24.95;
    item3 = 6.95;
    item4 = 14.95;
    item5 = 3.95;
    Tax = 0.06;
    Subtotal = item1 + item2 + item3 + item4 + item5;
    SalesTax = Subtotal * Tax;
    cout << "Price of item1 = $" << item1
    << "\nPrice of item2 = $" << item2
    << "\nPrice of item3 = $" << item3
    << "\nPrice of item4 = $" << item4
    << "\nPrice of item5 = $" << item5
    << "\nThe subtotal of the sale is " << Subtotal
    << "\nThe Sales Tax is " << SalesTax
    << "\nTotal = " << Subtotal + SalesTax;
    return 0;
}
```



BONUS TASK

Write a program that inputs a five-digit integer, separates the integer into its individual digits and prints the digits separated from one another by three spaces each.

Program:

```
Lab Task Bon...  
/storage/emulate...  
1 #include <iostream>  
2 using namespace std;  
3 int main()  
4     int num;  
5     cout<<"Enter 5 Digits";  
6     cin>>num;  
7  
8     int a= (num/10000);  
9     int b= (num/1000)%10;  
10    int c= (num/100)%10;  
11    int d= (num/10)%10;  
12    int e= num%10;  
13    cout<<a<<" "<<b<<" "<<c<<" "<<d<<" "  
14    "<<e;  
15    return 0;  
16 }
```

Output:

A terminal window with a dark background. The title bar is dark gray and contains a back arrow icon on the left, the text 'TAB' in the center, and a horizontal line and a vertical ellipsis icon on the right. The terminal content is white text on a black background. It shows the prompt 'Enter 5 Digits' followed by the input '12345'. The next line shows the output '1 2 3 4 5'. The final line shows '[Program finished]' followed by a cursor block.

```
Enter 5 Digits12345
1 2 3 4 5
[Program finished]
```

BONUS TASK

Write a program that inputs a five-digit integer, separates the integer into its individual digits separated from one another by three spaces each.

Program:

```
#include <iostream>
using namespace std;
int main(){
    int num;
    cout << "Enter five-digits : ";
    cin >> num;

    int a = (num / 10000);
    int b = (num / 1000) % 10;
    int c = (num / 100) % 10;
    int d = (num / 10) % 10;
    int e = (num % 10);
    cout << a << "   " << b << "   " << c << "   " <<
    d << "   " << e;
    return 0;
}
```

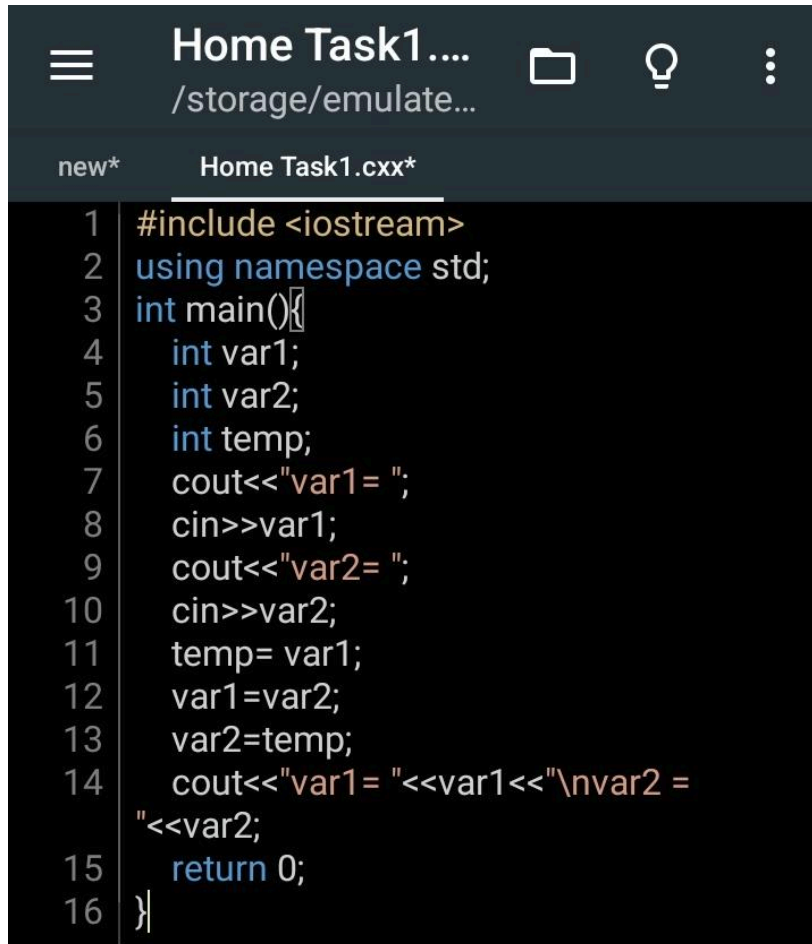
Output

```
Enter five-digits: 12345
1   2   3   4   5
```

Home Task 1: Swapping of Two Variables

Write a program that reads in two variables and swaps the value of these variables. For example if a variable 'var1' contains 10 and variable 'var2' contains 20, after the swap operation the variable 'var1' contains 20 (value of var2) and variable 'var2' contains 10 (value of var1).

Program:



```
Home Task1....  
/storage/emulate...  
new* Home Task1.cxx*  
1  #include <iostream>  
2  using namespace std;  
3  int main()  
4  {  
5      int var1;  
6      int var2;  
7      int temp;  
8      cout<<"var1 = ";  
9      cin>>var1;  
10     cout<<"var2 = ";  
11     cin>>var2;  
12     temp= var1;  
13     var1=var2;  
14     var2=temp;  
15     cout<<"var1 = "<<var1<<"\nvar2 = "  
16     "<<var2;  
17     return 0;  
18 }
```

Output:



```
← TAB _ ⋮  
var1= 10  
var2= 30  
var1= 30  
var2 = 10  
[Program finished]
```

Explanation:

We used a third temporary variable `temp` to store the value. First we save `var1` into `temp`, then we copy `var2` into `var1`, and finally we copy `temp` (original value of var1) into `var2`. In this way the values get exchanged.

Here is *Home Task 2* in same document style:

HOME TASK - 01

Write a program that reads in two variables and swaps the value of these variables. For

Example if a variable 'var1' contains 10 and 'var2' contains 20. After swap variable 'var1' contains 20 and 'var 2' contains 10 (value of var1).

Program:

```
#include <iostream>
using namespace std;
int main(){
    int var1;
    int var2;
    int temp;
    cout << "var1 = ";
    cin >> var1;
    cout << "var2 = ";
    cin >> var2;
    temp = var1;
    var1 = var2;
    var2 = temp;
    cout << "var1 = " << var1 << " \n var2 = " << var2;
    return 0;
}
```

Output

```
var 1 = 10
var 2 = 30
var 1 = 30
var 2 = 10
```

Explanation

We used third temporary variable to store the value.

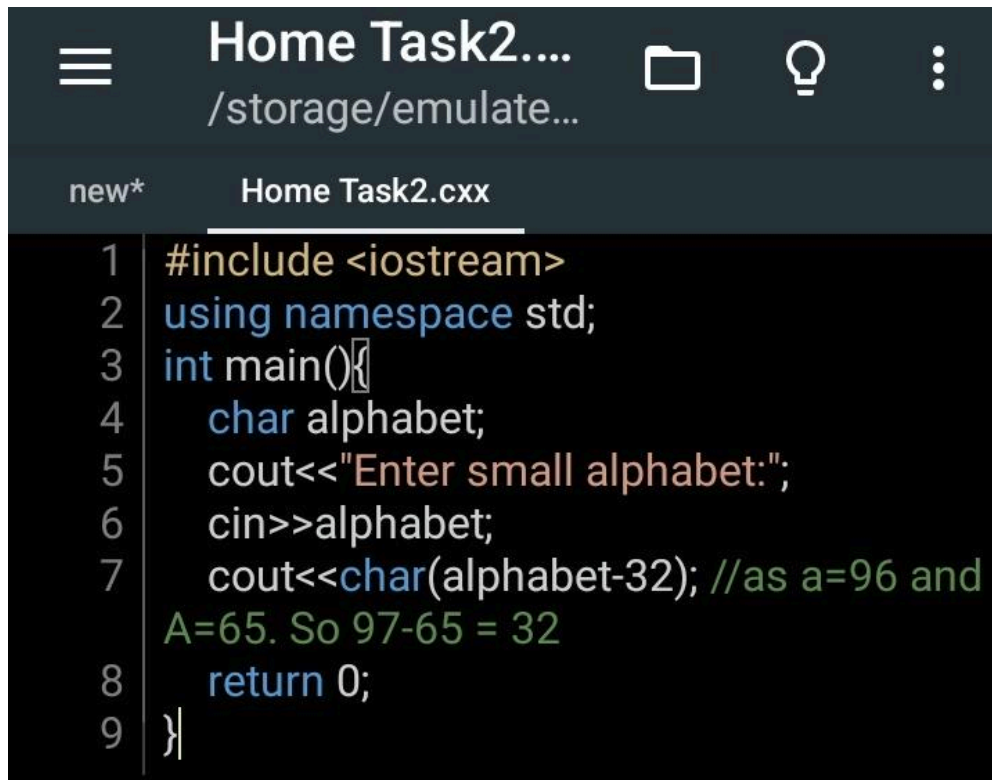
First we save var1 in temp, then copy var2 in var1, the temp (value of var1) in var2.

```
temp = var1
var1 = var2
var2 = temp
```

Home Task 2: Small to Capital Letter Conversion

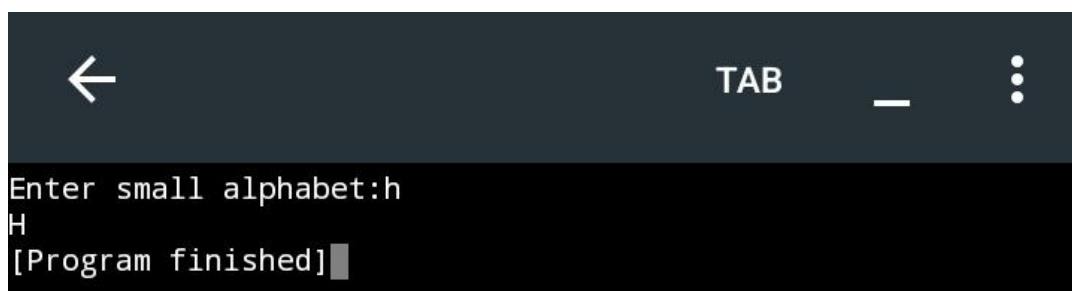
Make Upper Write a program that prompts users to enter the alphabet in small caps (a, b, c ... z) and display the entered alphabet into its upper caps (A, B, C ... Z). (hint ASCII)

Program:



```
Home Task2....  
/storage/emulate...  
new* Home Task2.cxx  
1 #include <iostream>  
2 using namespace std;  
3 int main(){  
4     char alphabet;  
5     cout<<"Enter small alphabet:";  
6     cin>>alphabet;  
7     cout<<char(alphabet-32); //as a=96 and  
   A=65. So 97-65 = 32  
8     return 0;  
9 }
```

Output:



```
← TAB _ ⋮  
Enter small alphabet:h  
H  
[Program finished]
```

Explanation:

In the ASCII table small letters have code 97 to 122 and capital letters have code 65 to 90. The difference is 32. So by subtracting 32 from small letters we get capital letters. For example 'b' is 98 - 32 = 66 which is 'B'.

HOME TASK - 02

Make UpperCase letters. Write a program that prompts users to enter the alphabet in small caps (a,b,c.....z) and display the entered alphabet into its uppercase. (A, B, C,Z) (Hint: ASCII)

Program:

```
#include <iostream>
using namespace std;
int main(){
    char alphabet;
    cout << "Enter small alphabet:";
    cin >> alphabet;
    cout << char(alphabet - 32);
    return 0;
}
```

Output

```
Enter small alphabet:h
H
```

Explanation: In ASCII table, small letters have code 97 to 122 and capital letters have code 65 to 90. The difference is 32. So by subtracting 32 from small alphabets we get capital letters.

As, 'b' is 98

$$98 - 32 = '66'$$

which is 'B'.